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Amal Omrani

PhD in Applied Mathematics

Research profile

Applied probability and mathematical modelling for risk management under market frictions and model uncertainty, combining stochastic analysis, convex duality and dynamic programming with numerical methods, open-source software and real data. Next direction: scenario-based modelling of climate-related risks, such as heat-driven electricity shortages, and green finance. Broader interests: stochastic control, Monte Carlo methods, uncertainty quantification, probabilistic machine learning.

Keywords Mathematical modelling; risk management; super-hedging; transaction costs; dynamic programming; convex duality; climate risk

Positions

2026–2027 **ATER** (Temporary Lecturer and Researcher), Université Paris Dauphine-PSL, Dec 2026–Aug 2027

2023–2026 **Doctoral researcher** with teaching duties, CEREMADE, Université Paris Dauphine-PSL

2019–2023 **Professeure agrégée**, Preparatory Institute for Engineering Studies (IPEIM), Monastir, Tunisia

Education

2023–2026 **PhD in Applied Mathematics**, Université Paris Dauphine-PSL, CEREMADE
Explicit Computation of Super-Hedging Prices and Strategies under Transaction Costs
Advisor: Emmanuel Lepinette. Defence: **17 November 2026**.
Main contributions: explicit super-hedging prices and strategies in discrete time, validated against exact benchmarks and on SPY data (2017–2025); a dynamic programming principle in Kabanov's solvency-cone model.
Jury: M. Rosenbaum (president), G. Di Nunno and L. Campi (referees), Y. Kabanov, E. Hubert.

2022–2023 **Master in Mathematics and Applications**, Université de Monastir (**ranked 1st**)
Thesis at LIMOS, Université Clermont Auvergne: reduced-complexity Gaussian processes for regression and classification (C. Samir, A. Fradi)

2019 **Agrégation in Mathematics** (Probability and Statistics), ENS Tunis (**ranked 3rd**)

Publications

[1] Lepinette, E., & **Omrani, A.** (2025). Explicit recursive construction of super-replication prices under proportional transaction costs. *Stochastics*, under revision (revised version submitted, October 2026). hal-05347450, arXiv:2503.02419

[2] Ben Amor, M. A., & **Omrani, A.** (2023). Chernoff's inequality in Riesz spaces. *Quaestiones Mathematicae*, *46*(9), 1777–1787. doi:10.2989/16073606.2022.2113474

[3] Lepinette, E., & **Omrani, A.** (2026). Computation of the super-hedging strategies for vanilla options with transaction costs. *Mathematics and Financial Economics*, under revision. hal-05626560

[4] Lepinette, E., & **Omrani, A.** (2026). Conditional random sets in Polish spaces with applications to the Skorokhod space. *Journal of Mathematical Analysis and Applications*, under revision. hal-05623197

[5] **Omrani, A.**, Fradi, A., & Samir, C. (2024). Reduced run-time and memory complexity regression with a Gaussian process prior. Conference paper, *55es Journées de Statistique (JDS)*, Bordeaux. jds2024.sciencesconf.org/528096

Manuscripts in preparation

[6] Lepinette, E., **Omrani, A.**, & Yan, W. Continuous-time markets with transaction costs: superhedging, dynamic programming, and path-dependent HJB–QVI.

[7] Ben Amor, M. A., & **Omrani, A.** Order-valued Bennett and Hoeffding inequalities in Riesz spaces.

Software

superhedge (with W. Yan, 2026): Python library for pricing and hedging options under transaction costs from bounded returns only: minimal capital, admissible hedges, calibration, backtesting. GitHub

Code for [1]: explicit recursion and optimal strategies; SPY backtest and sensitivity study. GitHub

Code for [3]: empirical SPY study, 2020–2026 (GitHub); strategies for calls and puts (GitHub)

Code for [5] (with A. Fradi): Gaussian-process regression with clustering. GitHub

Teaching

2026–2027	**Monte Carlo Methods** (Master 1), Paris Dauphine-PSL: 2 computer-lab groups in R and 2 tutorial groups; materials: sites.google.com/view/monte-carlo-methods
Spring 2027	**Statistics** (Bachelor L3), Paris Dauphine-PSL (scheduled)
2025–2026	**Introduction to Finance** (Bachelor), Paris Dauphine-PSL: derivatives, options (30 students, 40h)
2024–2025	**Statistics** (Bachelor L2), Paris Dauphine-PSL: lectures, tutorials and assessments (30 students, 50h)
2024–2025	**AI Projects**, Paris Dauphine-PSL: supervision of data-analysis projects in Python (30 students, 15h)
2019–2023	**Algebra I & II**, IPEIM Monastir: full responsibility for lectures, tutorials and exams (60 students/year, 18h/week); grader for the national engineering-school entrance exams

Supervision

2026–2027	Master's thesis: *Monte Carlo methods for PDEs: random walks and the Dirichlet problem*

Talks

Oct 2026	FiME PhD Seminar, Institut Henri Poincaré	Paris, France
Sept 2026	COSA 2026, CIRM	Marseille, France
June 2026	13th Bachelier World Congress	Bologna, Italy
Mar 2026	CEREMADE Third-Year PhD Day	Paris, France
Oct 2025	Stochastics in Mathematical Finance and Physics (SMFP 2025)	Hammamet, Tunisia
Sept 2025	Stochastics & Computational Finance (SCF 2025)	Lisbon, Portugal
June 2025	Young Researchers' Days, CEREMADE	Paris, France
Apr 2025	XXVI Workshop on Quantitative Finance (QFW 2025)	Palermo, Italy
Jan 2025	17th Bachelier Colloquium	Métabief, France
May 2024	55es Journées de Statistique (JDS 2024)	Bordeaux, France
Oct 2022	FAOTSA22: Functional Analysis and Operator Theory Workshop	Kruger Park, South Africa

Schools and seminars attended

Jan 2026	23rd Winter School on Mathematical Finance	Soesterberg, Netherlands
July 2025	5th Barcelona Summer School of Stochastic Analysis and Quantitative Finance	Barcelona, Spain
June 2024	CIMPA School: Ordered structures in finance and machine learning	Sousse, Tunisia
May 2022	Workshop on Banach Spaces and Banach Lattices II, ICMAT	Madrid, Spain
Regular	Bachelier Seminar	Paris, France

Service

2026, 2023	Co-organizer, COSA 2026 (CIRM, Marseille; Natixis Foundation support) and COSA 2023 (Tozeur, Tunisia)
2024–2025	Co-organizer, Young Researchers' Days, CEREMADE

Skills

Computing	Python (NumPy, SciPy, LP solvers), R, C++, MATLAB, Git, LaTeX
Languages	English (fluent), French (fluent), Arabic (native), Italian (basic)